

Windows Authentication Investigation

Project 3 — Controlled Security Log Analysis

Investigation Type	Windows Security Log Analysis
Platform	Windows 10
Primary Event IDs	4625 (failed logon), 4624 (successful logon)
Test Account	LabTest
Activity Date	August 19, 2026
Status	Controlled lab exercise

1. Executive Summary

A controlled Windows authentication investigation was performed using Event Viewer. Five failed logon events (Event ID 4625) were generated against a dedicated test account, followed later by a successful logon (Event ID 4624). The events were correlated by account, time, logon type, and source information. The evidence showed local interactive authentication from 127.0.0.1. Because the activity was intentionally generated for this lab, no real compromise or remote brute-force attack is claimed.

2. Investigation Objectives

- Identify failed authentication events in the Windows Security log.
- Examine Event ID 4625 details and authentication failure codes.
- Correlate repeated failures with a later successful Event ID 4624.
- Determine whether the activity originated locally or remotely.
- Document an evidence-based analyst conclusion.

3. Evidence Collected

Time	Event ID	Account	Result
10:50:03	4625	LabTest	Failed
10:50:10	4625	LabTest	Failed
10:50:17	4625	LabTest	Failed
10:50:22	4625	LabTest	Failed
10:50:31	4625	LabTest	Failed
10:58:20	4624	LabTest	Successful

The five failed attempts occurred within approximately 28 seconds. The later successful logon occurred approximately eight minutes after the final failed attempt.

4. Technical Analysis

Event ID 4625: The failed logon records targeted the LabTest account. The events showed Logon Type 2 (Interactive), Status 0xC000006D, SubStatus 0xC000006A, and the Negotiate authentication package. The source address was 127.0.0.1 with port 0.

Event ID 4624: The successful logon targeted LabTest and showed Logon Type 2 (Interactive), Negotiate authentication, and 127.0.0.1 as the source address.

Source assessment: 127.0.0.1 is the local loopback address. Combined with the interactive logon type, this indicates the observed authentication activity was local to the Windows system.

5. Analyst Assessment

The event pattern is consistent with repeated password-entry failures followed by a successful authentication. In a production environment, repeated failures followed by success could warrant additional investigation. In this lab, however, the attempts were intentionally generated against a test account. Therefore, the activity should be classified as controlled lab activity rather than a confirmed security incident.

6. Recommendations

- Continue monitoring Event IDs 4625 and 4624 for unusual authentication patterns.
- Correlate failed logons with source IP, workstation, account, and logon type.
- Investigate repeated failures followed by a success when the source or account context is unexpected.
- Use account lockout policies and strong authentication controls in production environments.
- Preserve relevant event records before making severity or compromise determinations.

7. Lessons Learned

This exercise demonstrates why security analysts should validate the complete event context before declaring an attack. Event frequency alone is not sufficient: source address, logon type, account, timing, and known lab conditions all affect the final assessment.

8. Evidence Screenshots

Figure 1 — Sanitized Event ID 4625 evidence.

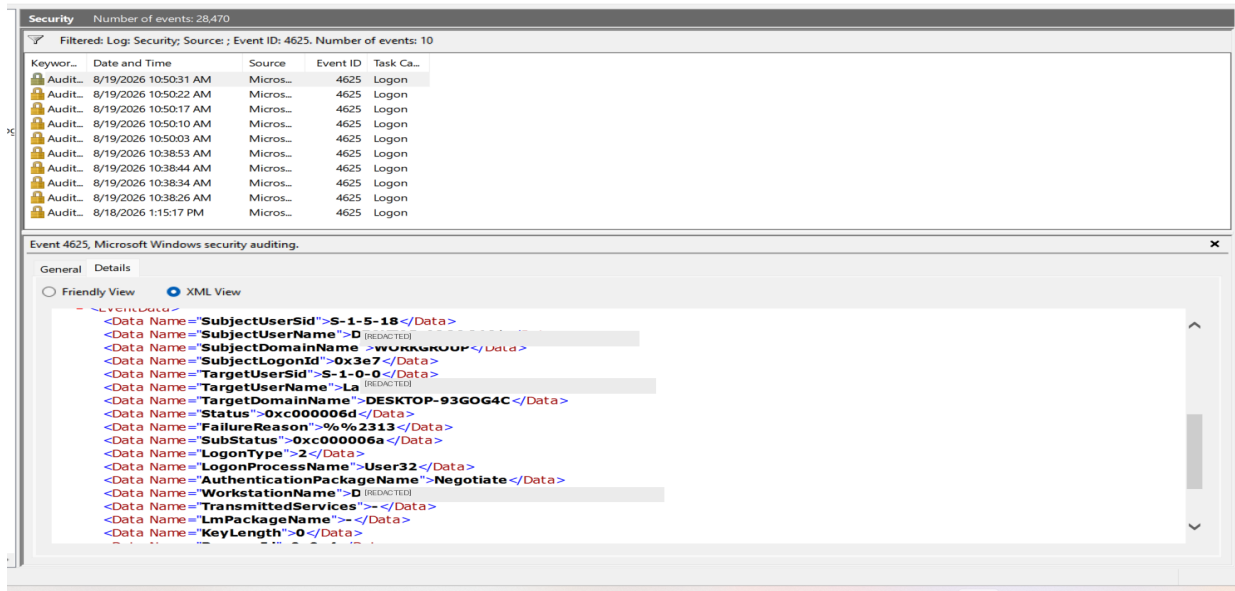


Figure 2 — Sanitized Event ID 4624 XML details.

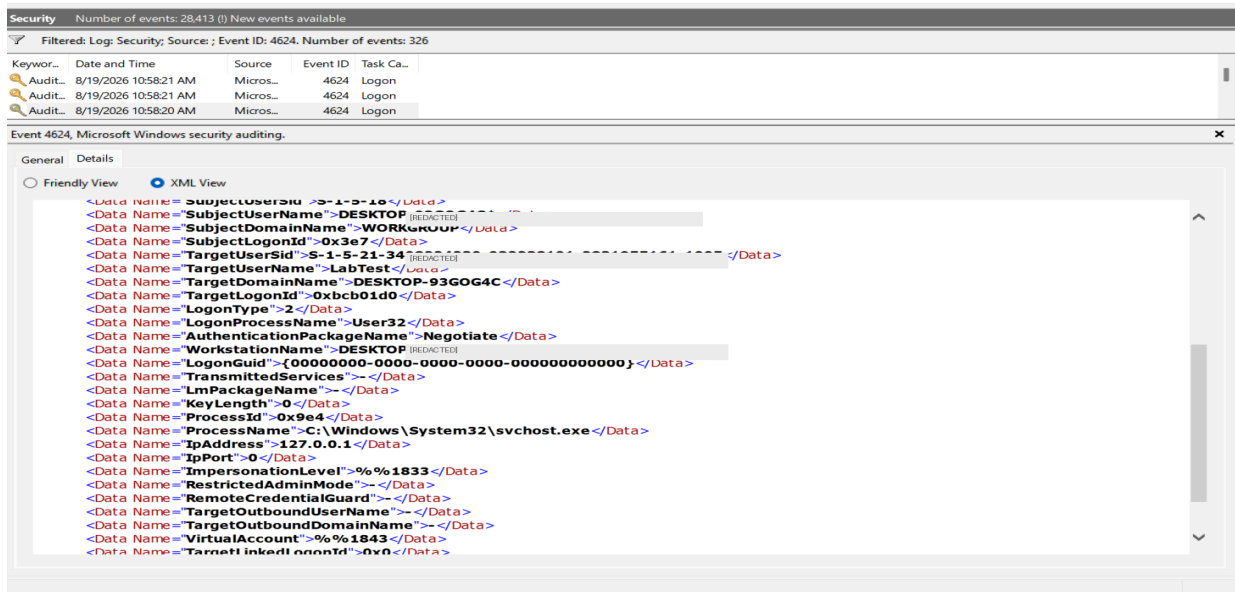


Figure 3 — Sanitized Event ID 4624 general summary.

Security

Number of events: 28413 (0) New events available

Filtered: Log: Security; Source: ; Event ID: 4624. Number of events: 326

Keywords	Date and Time	Source	Event ID	Task Category
Audit...	8/19/2026 10:58:21 AM	Micros...	4624	Logon
Audit...	8/19/2026 10:58:21 AM	Micros...	4624	Logon
Audit...	8/19/2026 10:58:20 AM	Micros...	4624	Logon

Event 4624, Microsoft Windows security auditing.

General

Details

An account was successfully logged on.

Subject:

Security ID: [REDACTED]

Account Name: [REDACTED]

Account Domain: WORKGROUP

Logon ID: 0x3E7

Logon Information:

Logon Type: 2

Restricted Admin Mode: -

Remote Credential Guard: -

Virtual Account: No

Elevated Token: No

Impersonation Level: Impersonation

New Logon:

Security ID: [REDACTED]

Account Name: [REDACTED]

Account Domain: DESKTOP-93GOG4C

Log Name: Security

Source: Microsoft Windows security auditing

Event ID: 4624

Level: Information

User: N/A

OpCode: Info

Task Category: Logon

Keywords: Audit Success

Computer: DESKTOP-93GOG4C

8/19/2026 10:58:20 AM